

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Municipal Solid Waste Route Optimizer (MSWRO)

Item	Details
Objective	Convert MSWRO functional requirements into Agile Epics and User Stories, estimate them, plan sprints, simulate Scrum
Jira workspace	MSWRO board
Epics	4 functional Epics
User Stories	17
Total Story Points	66
Sprint 1	5 stories — 24 Story Points
Sprint 2	8 stories — 42 Story Points
Workflow demonstrated	To Do → In Progress → Done

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This report documents the Jira backlog, story-point estimation, two sprint plans, progressive story completion, burndown/velocity evidence, and reflection on the Scrum simulation.

1. EPICS AND USER STORIES

User Stories use the Agile convention: **As a [role], I want [goal], so that [benefit]**. Story Points use Fibonacci-style relative sizing.

Epic 1 — Smart Bin & Telemetry Management

Key	User Story	Priority	SP
MSWRO-5	Ingest ultrasonic bin fill-level data every 15 minutes for route planning.	High	5
MSWRO-6	Update bin status in real time so route decisions use current information.	High	3
MSWRO-7	Maintain reliable telemetry ingestion for continuous operational data.	High	8
MSWRO-8	Identify bins exceeding the 80% fill threshold for priority collection.	High	3

Epic total: 19 Story Points

Epic 2 — Route Optimization & Dispatch

Key	User Story	Priority	SP
MSWRO-9	Generate optimized daily collection routes for efficient vehicle movement.	High	8
MSWRO-10	Display route metrics and a route map for efficiency evaluation.	High	5
MSWRO-11	Recalculate routes within 45 seconds when conditions change.	High	8

Epic total: 21 Story Points

2. REMAINING EPICS AND STORY-POINT SUMMARY

Epic 3 — Driver Collection Operations

Key	User Story	Priority	SP
MSWRO-12	Provide turn-by-turn navigation for the assigned collection route.	High	5
MSWRO-13	Mark a bin as collected so completed pickups are recorded accurately.	High	3
MSWRO-14	Mark a bin as blocked or inaccessible and record the reason.	High	3

Epic total: 11 Story Points

Epic 4 — Citizen Complaints & Supervisor Control

Key	User Story	Priority	SP
MSWRO-15	Submit a missed-pickup complaint so the issue can be recorded and acted upon.	High	5
MSWRO-16	Override bin collection status to correct exceptional or incorrect updates.	High	5
MSWRO-17	Reassign collection routes to drivers when operational changes occur.	High	5

Epic total: 15 Story Points

Epic	Stories	Story Points
Smart Bin & Telemetry Management	4	19
Route Optimization & Dispatch	3	21
Driver Collection Operations	3	11
Citizen Complaints & Supervisor Control	3	15
TOTAL	13	66

The backlog contains 17 User Stories overall. The Jira board uses the assigned Fibonacci-style points to represent relative effort, complexity, risk and uncertainty.

3. SPRINT 1

Sprint Goal: Complete the core smart-bin monitoring and initial route/driver workflow needed to support route planning and field collection.

Story	Summary	SP	Flow
MSWRO-5	Ingest ultrasonic bin fill-level data	5	To Do → In Progress → Done
MSWRO-6	Update bin status in real time	3	To Do → In Progress → Done
MSWRO-8	Identify bins exceeding 80% fill threshold	3	To Do → In Progress → Done
MSWRO-9	Generate optimized daily collection routes	8	To Do → In Progress → Done
MSWRO-12	Provide turn-by-turn navigation	5	To Do → In Progress → Done
TOTAL		24	

Figure 1. Jira backlog showing MSWRO Sprint 1 populated with five stories and 24 Story Points.

4. SPRINT 2

Sprint Goal: Complete the remaining route optimization, driver collection, and citizen/supervisor control functions.

Story	Summary	SP
MSWRO-7	Maintain reliable telemetry ingestion	8
MSWRO-17	Reassign collection routes to drivers	5
MSWRO-14	Mark bin as blocked or inaccessible	3
MSWRO-13	Mark bin as collected	3
MSWRO-16	Override bin collection status	5
MSWRO-15	Submit missed-pickup complaint	5
MSWRO-11	Recalculate routes within 45 seconds	8
MSWRO-10	Display route metrics and map	5
TOTAL		42

The screenshot shows a Jira Sprint board for 'MSWRO Sprint 2' (15 Sep – 28 Sep, 8 work items). The board displays eight selected stories, each with a checkbox, ID, summary, a progress bar, a status dropdown, a story point value, and an assignee icon. The total story points for the selected items are 42. The stories are:

- ☒ MSWRO-7 Maintain reliable telemetry ingestion (Smart Bin..., To Do, 8)
- ☒ MSWRO-17 Reassign collection routes to drivers (Citizen C..., To Do, 5)
- ☒ MSWRO-14 Mark bin as blocked or inaccessible (Driver Col..., To Do, 3)
- ☒ MSWRO-13 Mark bin as collected (Driver Col..., To Do, 3)
- ☒ MSWRO-16 Override bin collection status (Citizen C..., To Do, 5)
- ☐ ☒ MSWRO-15 Submit missed-pickup complaint (Citizen C..., To Do, 5)
- ☒ MSWRO-11 Recalculate routes within 45 seconds (Route Opt..., To Do, 8)
- ☒ MSWRO-10 Display route metrics and map (Route Opt..., To Do, 5)

Figure 2. Jira Sprint 2 showing eight selected stories totaling 42 Story Points.

5. SPRINT EXECUTION AND STATUS PROGRESSION

The sprint simulation demonstrated the required Scrum status flow. Stories were initially placed in **To Do**, then progressed through **In Progress**, and finally moved to **Done**. The sprint was also reopened so the status progression could be demonstrated again.

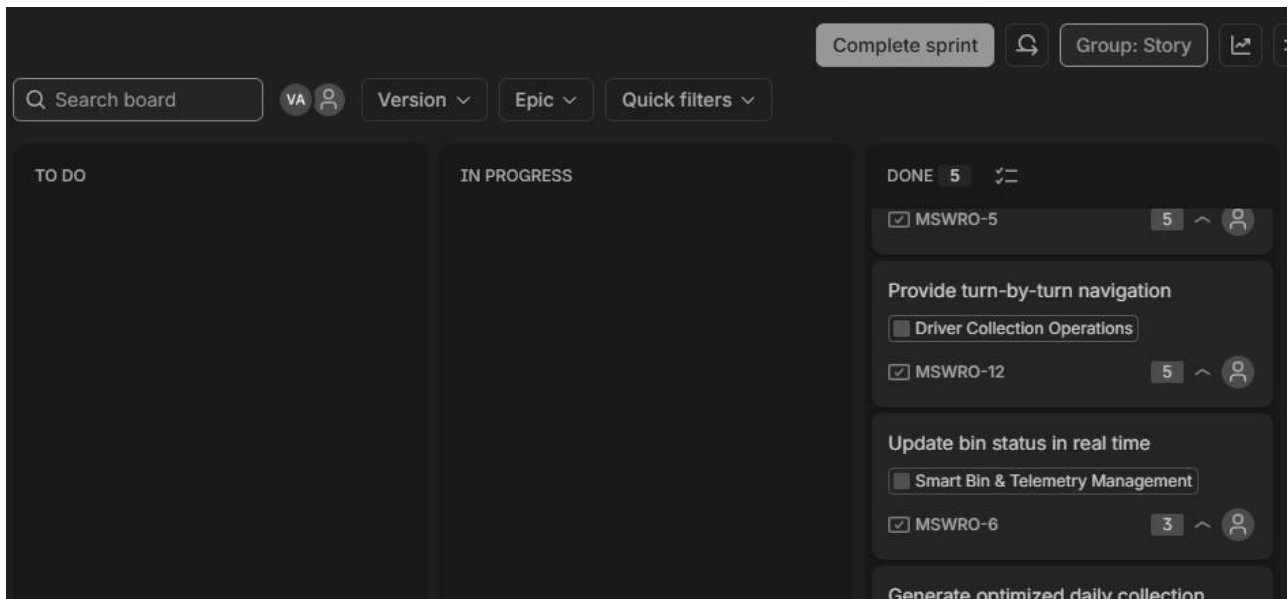


Figure 3. Jira Active Sprint evidence showing completed stories in the Done column.

6. BURNDOWN CHART EVIDENCE

The Burndown Chart tracks remaining Story Points against the planned guideline and identifies non-working-day periods. The Sprint 1 chart provides the clearest visual evidence of work decreasing from the initial 24 points to zero.

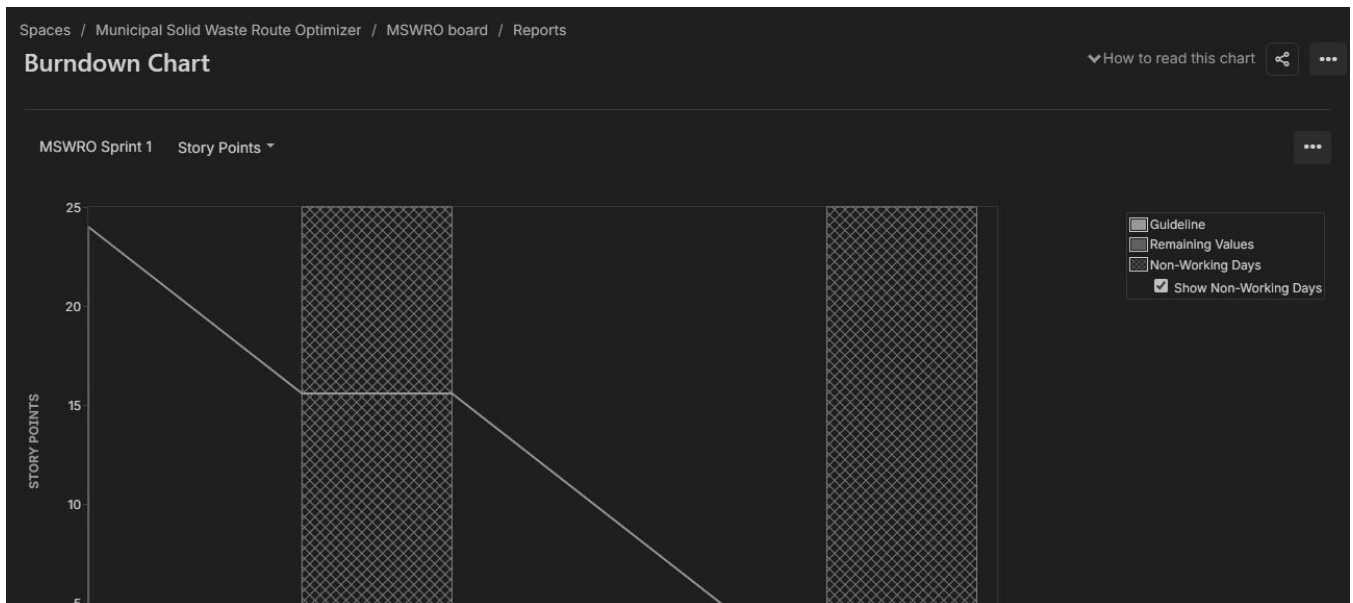


Figure 5. Jira Burndown Chart for MSWRO Sprint 1.



Figure 6. Jira Burndown Chart for MSWRO Sprint 2. In the captured simulation, the remaining-values line is at zero; the chart scale is compressed in this view.

7. VELOCITY AND SPRINT REPORT EVIDENCE

The Velocity Report records the amount of work committed and completed in each displayed sprint. In the captured report, Sprint 1 shows 24 Story Points committed and completed, while the configured average line is 12 Story Points. Sprint 2's 42-point completion is separately evidenced by the Sprint Report.



Figure 7. Jira Velocity Report showing Sprint 1 commitment/completion of 24 Story Points and an average of 12.

Spaces / Municipal Solid Waste Route Optimizer / MSWRO board / Reports

Sprint Report

▼ How to read this chart [🔗](#) [...](#)

Status Report

TIME

Issues completed outside of this sprint [View in Issue Navigator](#)

Key	Summary	Issue Type	Priority	Status	Story Points (42)
MSWRO-7	Maintain reliable telemetry ingestion	☑ Story	^ High	DONE	8
MSWRO-10	Display route metrics and map	☑ Story	^ High	DONE	5
MSWRO-11	Recalculate routes within 45 seconds	☑ Story	^ High	DONE	8
MSWRO-13	Mark bin as collected	☑ Story	^ High	DONE	3
MSWRO-14	Mark bin as blocked or inaccessible	☑ Story	^ High	DONE	3
MSWRO-15	Submit missed-pickup complaint	☑ Story	^ High	DONE	5
MSWRO-16	Override bin collection status	☑ Story	^ High	DONE	5
MSWRO-17	Reassign collection routes to drivers	☑ Story	^ High	DONE	5

Figure 8. Jira Sprint Report listing the eight Sprint 2 stories completed, totaling 42 Story Points.

8. REFLECTION QUESTIONS

1. Did your estimations reflect the actual effort?

For the simulation, yes. Story Points provided a consistent relative estimate of effort. Sprint 1 contained 24 points and Sprint 2 contained 42 points, and the selected stories were ultimately completed. The estimates should be interpreted as relative sizing, not as direct hours of work.

2. Was your backlog well-prioritized?

Yes. The backlog was organized around four functional areas and the sprint selections emphasized High-priority stories. Sprint 1 focused on core telemetry, threshold detection, route generation and navigation. Sprint 2 covered telemetry reliability, route recalculation/display, field collection controls, and citizen/supervisor functions.

3. How did your simulated sprint align with your plan?

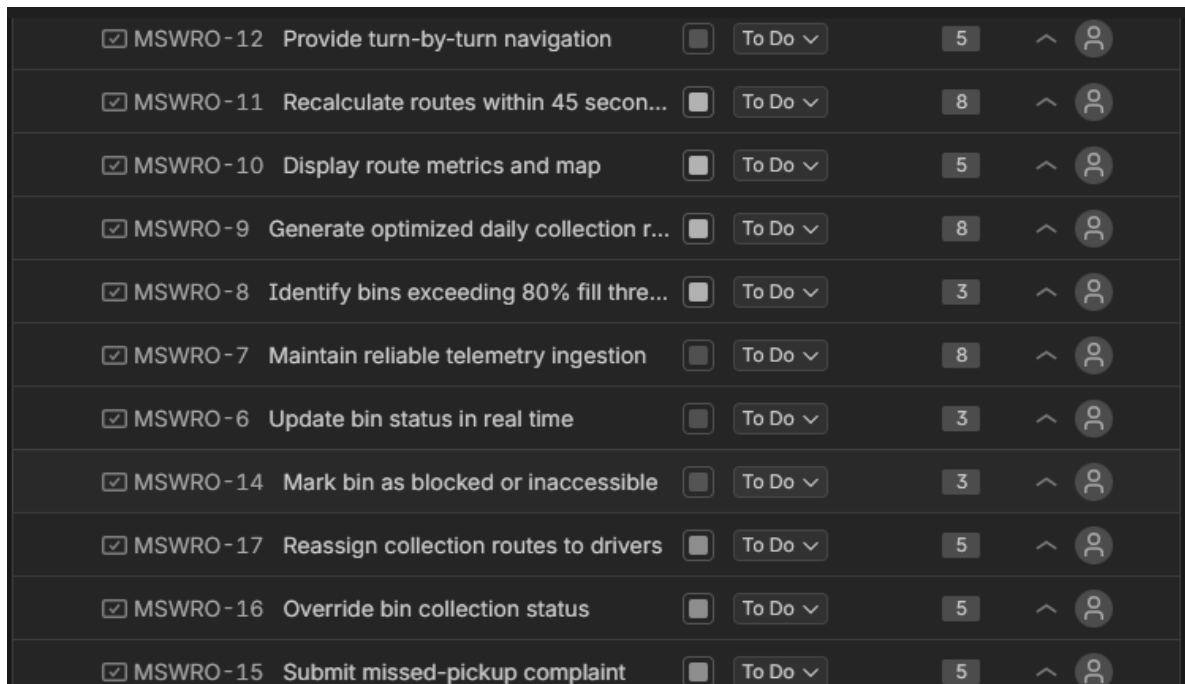
The selected work was planned into sprints and then progressed through To Do → In Progress → Done. Sprint 2 was reopened during the exercise and the stories were again progressed through the workflow, making the Scrum state transitions visible in Jira.

4. What insights did the burndown chart give about team capacity?

The burndown shows how remaining work changes relative to the ideal guideline. The simulation ultimately completed the planned sprint workloads: 24 Story Points in Sprint 1 and 42 in Sprint 2. Because this is a lab simulation rather than a long-running development project, these numbers should not be treated as a reliable long-term velocity forecast.

9. FINAL JIRA EVIDENCE AND CONCLUSION

The final Jira workspace contains the configured Epics, User Stories, story-point estimates, sprint plans, status transitions, and reporting evidence required for the Lab 2 exercise.



☒	MSWRO-12	Provide turn-by-turn navigation	☐ To Do	5	^	
☒	MSWRO-11	Recalculate routes within 45 secon...	☐ To Do	8	^	
☒	MSWRO-10	Display route metrics and map	☐ To Do	5	^	
☒	MSWRO-9	Generate optimized daily collection r...	☐ To Do	8	^	
☒	MSWRO-8	Identify bins exceeding 80% fill thre...	☐ To Do	3	^	
☒	MSWRO-7	Maintain reliable telemetry ingestion	☐ To Do	8	^	
☒	MSWRO-6	Update bin status in real time	☐ To Do	3	^	
☒	MSWRO-14	Mark bin as blocked or inaccessible	☐ To Do	3	^	
☒	MSWRO-17	Reassign collection routes to drivers	☐ To Do	5	^	
☒	MSWRO-16	Override bin collection status	☐ To Do	5	^	
☒	MSWRO-15	Submit missed-pickup complaint	☐ To Do	5	^	

Figure 9. Jira backlog evidence showing the complete set of MSWRO User Stories and story-point assignments.

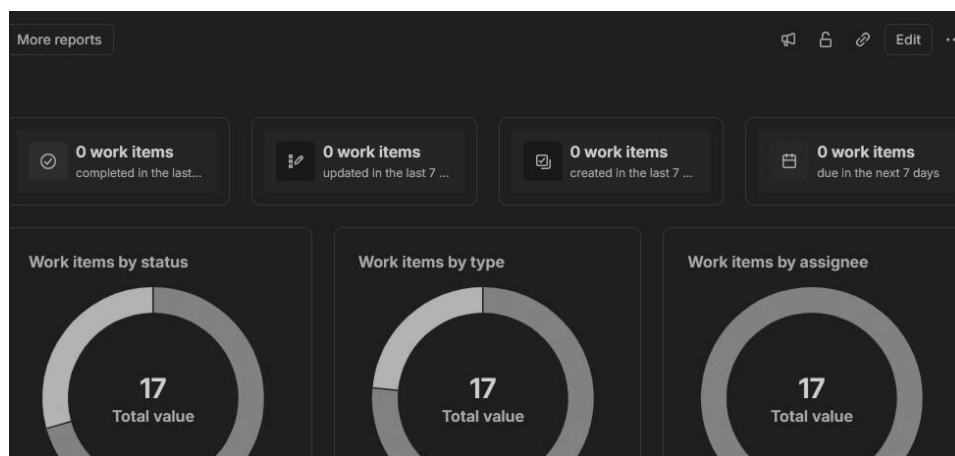


Figure 10. Jira Reports view providing project-level progress/reporting evidence.

Conclusion: The MSWRO Jira workspace demonstrates the complete Lab 2 Agile cycle: backlog creation, Epic/User Story organization, prioritization, Fibonacci-style Story Point estimation, sprint planning, progressive status movement, sprint completion, burndown/velocity analysis, and reflection.